

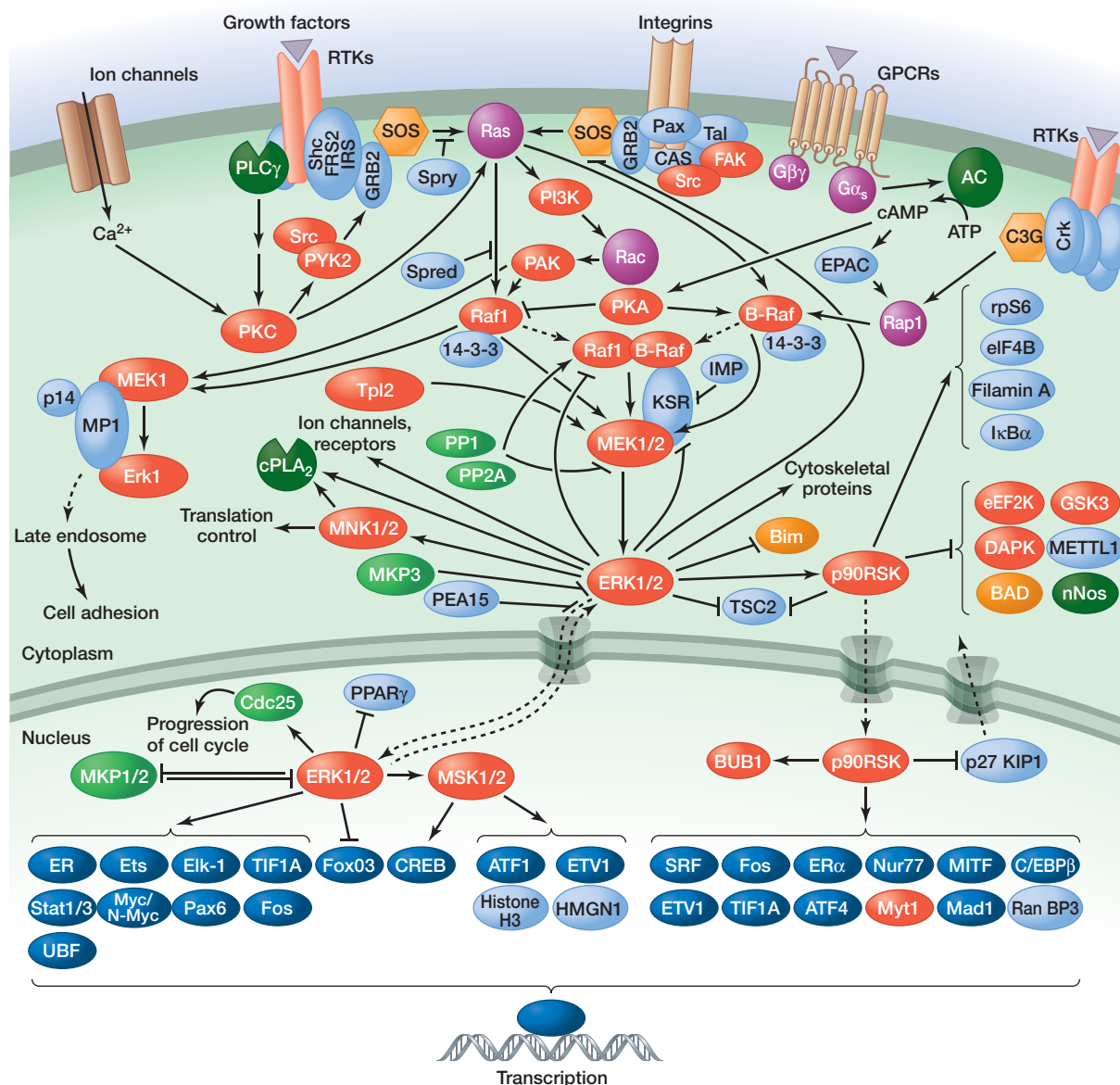
MAP Kinase Pathways

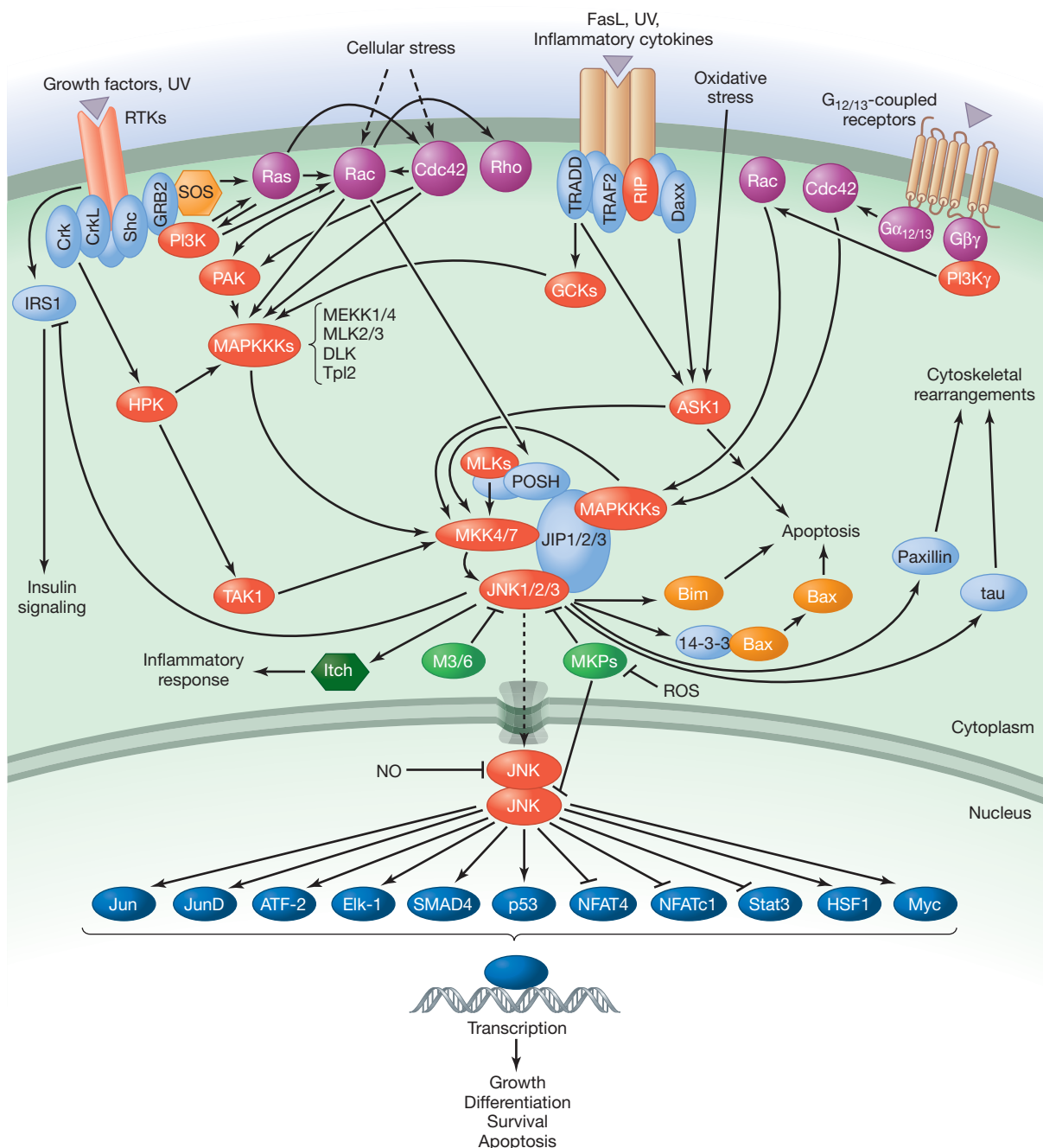
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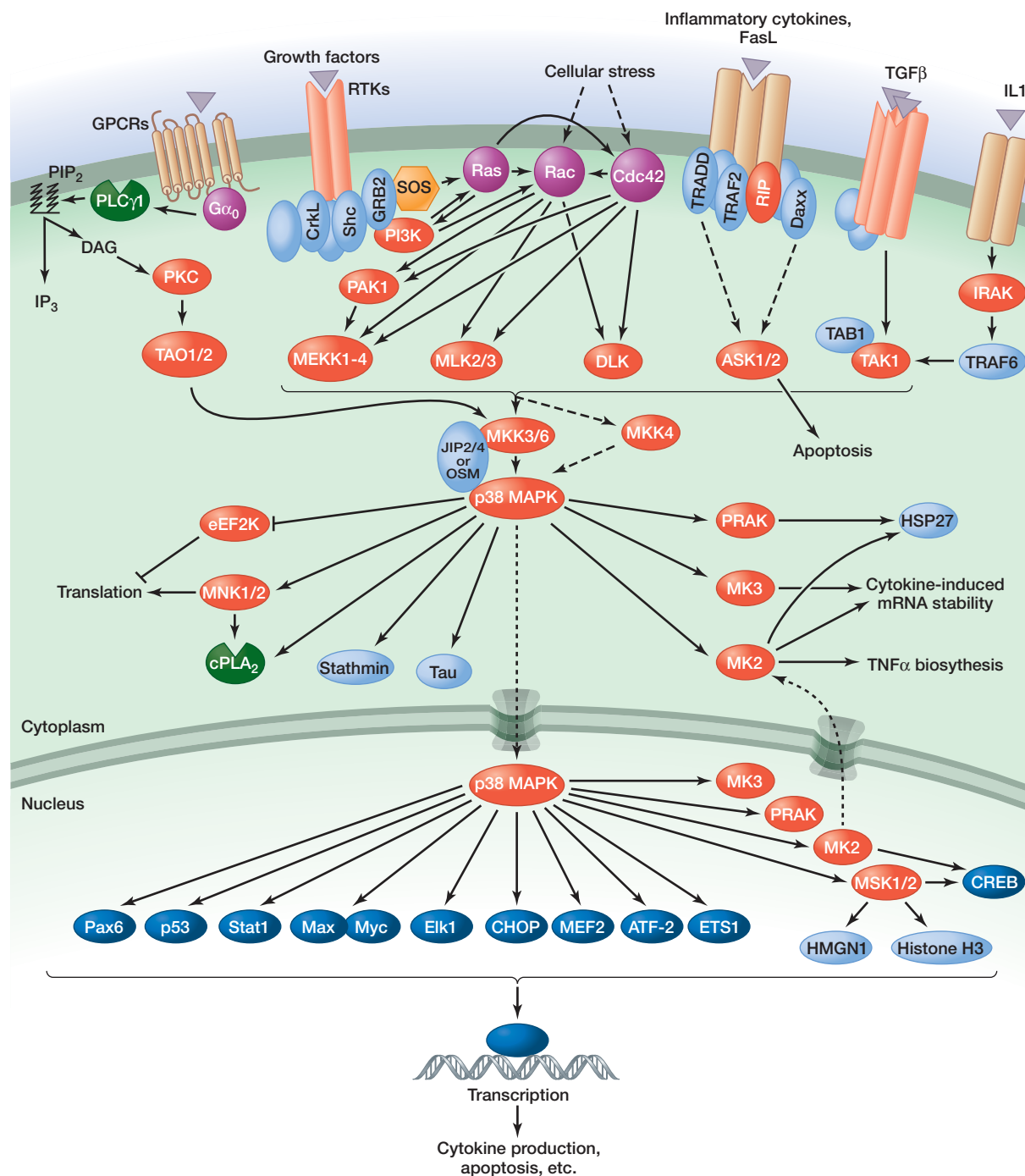
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Mitogen-activated protein kinase (MAPK) modules containing three sequentially activated protein kinases are key components of a series of vital signal transduction pathways that regulate processes such as cell proliferation, cell differentiation, and cell death in eukaryotes from yeast to humans (Fig. 1) (Qi and Elion 2005; Raman et al. 2007; Keshet and Seger 2010). Each cascade is initiated by specific extracellular cues and leads to activation of a particular MAPK following the successive activation of a MAPK kinase kinase (MAPKKK) and a MAPK kinase (MAPKK) (Fig. 1). The MAPKKK is typically activated by interactions with a small GTPase and/or phosphorylation by protein kinases downstream from cell surface receptors (Cuevas et al. 2007). The MAPKKK directly phosphorylates and activates the MAPKK, which, in turn, activates the MAPK by







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